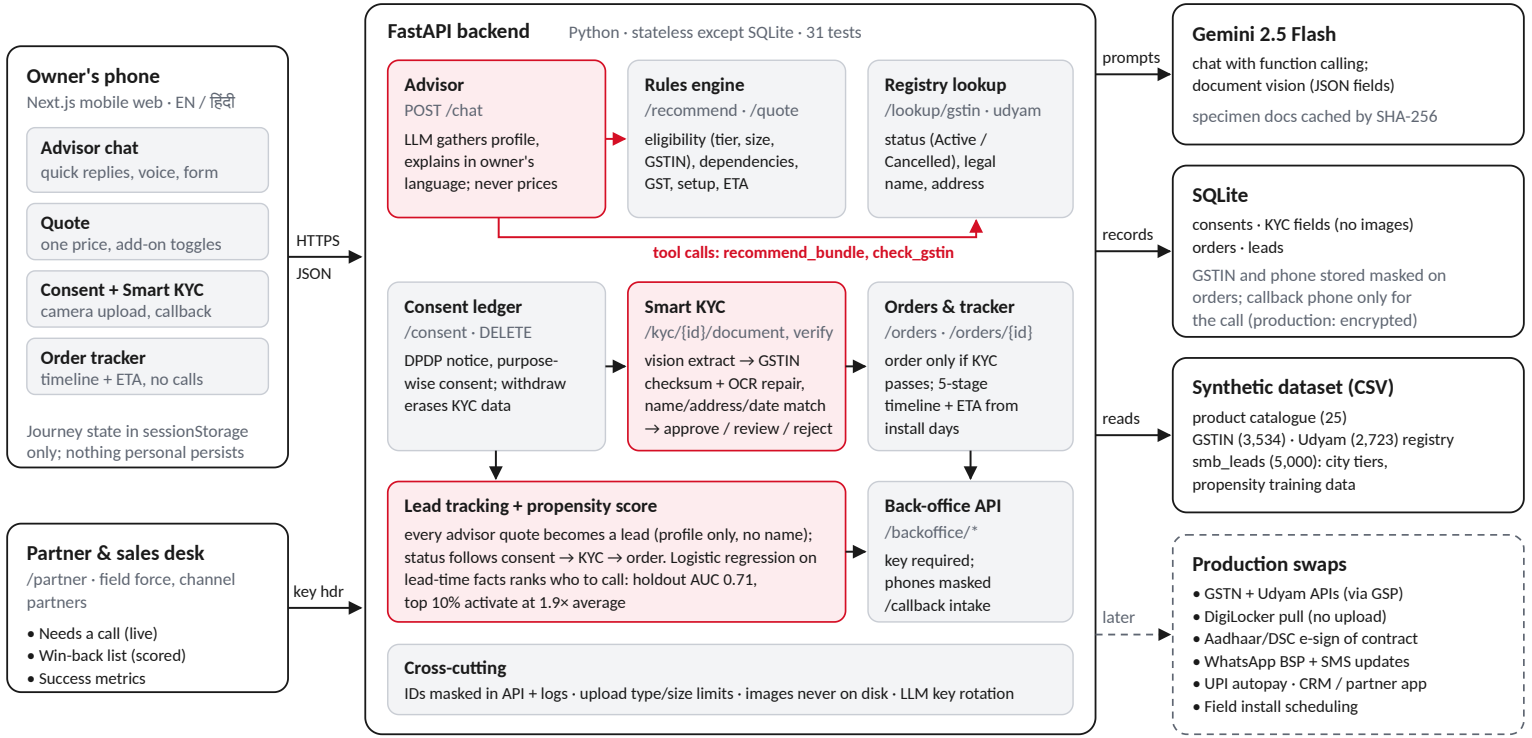
Solution architecture

Udyam Digital · MSME acquisition and onboarding for NorthStar Business · Airtel SYNC, Enterprise Track A

Solid boxes: built and running in the prototype
Dashed boxes: production integrations



Data flow through one journey

1. **Advise.** The owner chats (or fills the short form). Gemini asks a few questions and must call `recommend_bundle`; the rules engine picks eligible products and prices, so the LLM can explain but never invent a price. A lead row (profile only) is created.
2. **Quote.** `/quote` recomputes the monthly total, 18% GST, setup, contract and activation ETA live as add-ons are toggled; dependency errors block invalid carts.
3. **Consent.** The DPDP notice lists purposes, data collected and not collected, and retention. Required purposes are recorded before any document can be uploaded.
4. **Smart KYC.** Each upload is read by the vision model, the image is dropped, and only fields are kept. `verify` checks GSTIN format and check digit (with safe OCR repair), registry status and names, Udyam, signatory ID validity and bill age (≤ 90 days).
5. **Order and track.** An order needs KYC that is not rejected and an active GSTIN for GST-only products. The tracker shows 5 stages with expected dates, replacing status-chase calls.
6. **Hand-off.** Anything not closed (quote but no KYC, failed KYC, callback request) appears on the partner desk, ranked by propensity, with the next action.

Why it is built this way

- **LLM talks, rules decide.** Prices, eligibility and KYC verdicts are deterministic and unit-tested; the form fallback works if the LLM is down.
- **Minimal data.** No Aadhaar, PAN or bank details asked for; images never stored; identifiers masked in every response and log line.
- **Partners add, not compete.** The digital flow handles the easy 80%; partners get a ranked, reasoned call list instead of cold leads, and can send the WhatsApp advisor link to leads who never had time for a visit.
- **Cheap to run.** One small API, one database, pay-per-call LLM (4 calls per KYC). Scales out by moving SQLite to Postgres and the session store to Redis.

Measured on

Lead-to-activation 7.4% → 15% · days to activate 17.6 → ≤ 7 · cost per activation ₹15k → ≤ ₹6k · status-chase calls 3 → ≤ 0.5 · 6-month churn ≈ 15% → ≤ 10%.

All businesses, people and identifiers are synthetic (hackathon dataset). NorthStar Business stands in for the fictional operator in the case.